PLC IS GOOD !

PLC Fundamentals (4)

Objectives

- MLO #2.20: Describe the processes involved in the scan and what the scan time is.
- MLO #2.21: Determine the scan time diagram with and without feedback for simple ladder diagrams.
- MLO #2.22: Describe the A-B SLC 500 file organization by naming the files involved and the data they hold.

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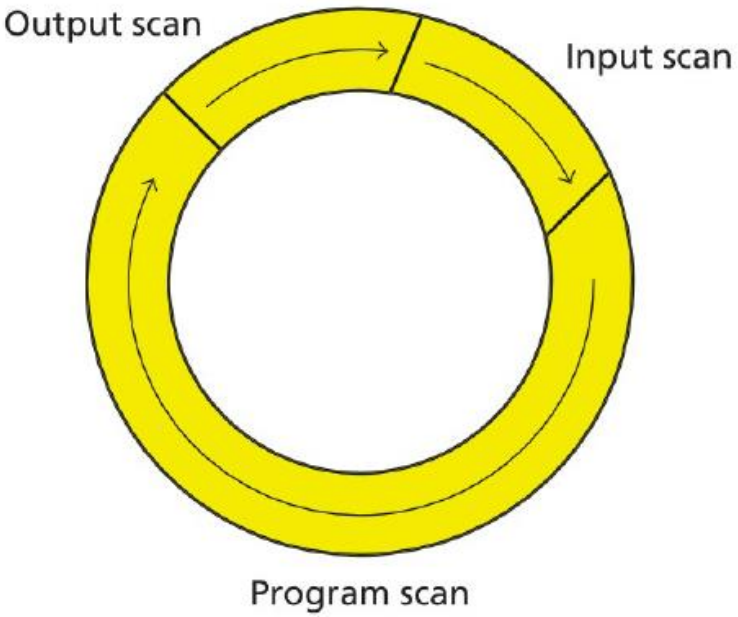
Scan Sequencing

Obtaining Time Diagrams

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Program Scan (Review)

- PLC program execution flow on each rung is from left to right
- From the top rung to the bottom rung, moving through the entire PLC ladder logic diagram
- Operational Scan Rate:
  - Time required to execute the PLC ladder diagram once



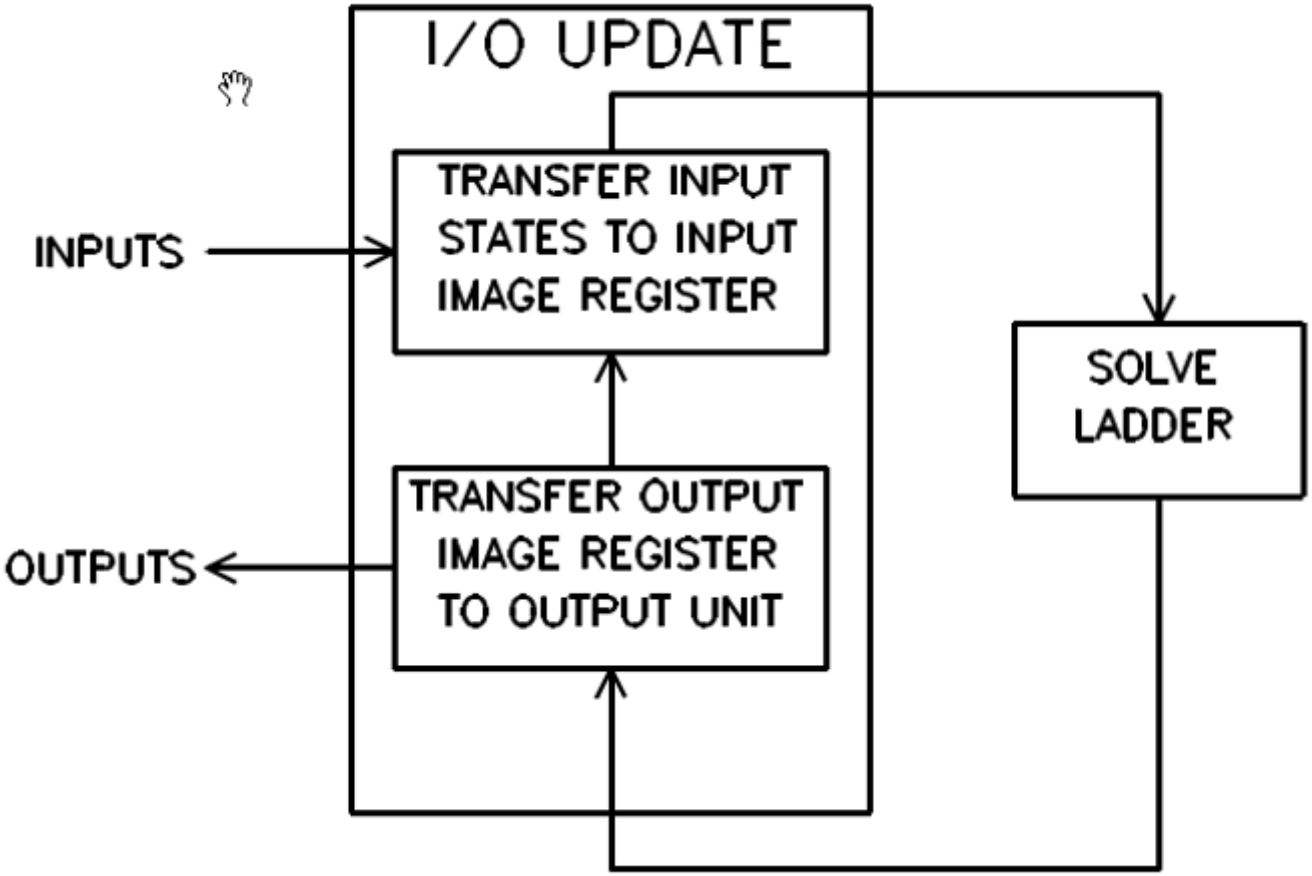
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PLC Operational Cycle

- Input scan:
  - Processor reads input ports and updates input status table
- Program scan:
  - Processor executes PLC program and updates output status table
- Output scan:
  - Output status table values transferred to output terminals

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Program Scan (Cont.)



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PLC Program Scan Time

- PLC ladder logic diagram instructions:
  - Scanned from instruction in the upper-left corner
  - Ends with instruction on the lower-right corner
  - After last instruction in the lower-right corner, scanning restarts
  - Instructions are done from left to right and
  - Top - down

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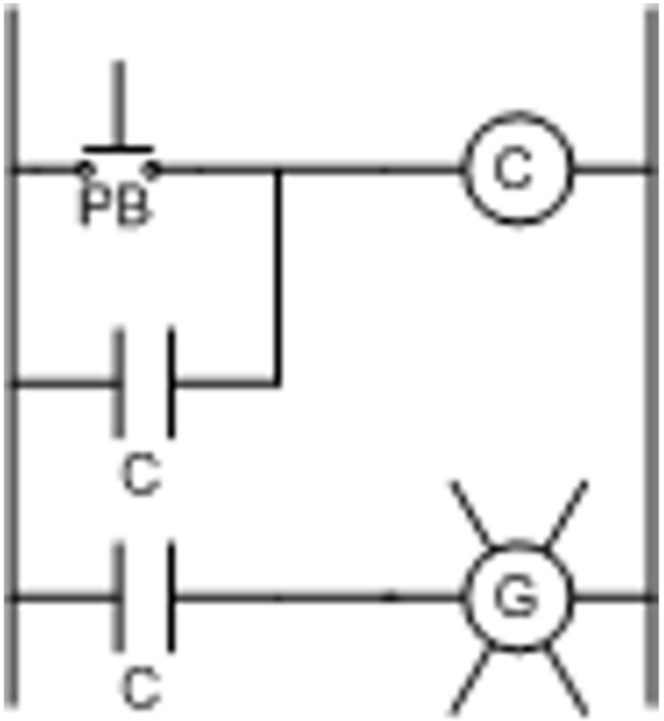
Scanning a Rung

- Input and output addresses are updated. Inputs are updated at the beginning of the scan, outputs are updated at the end of the scan.
- Sometimes, specific inputs or outputs may not be updated in time

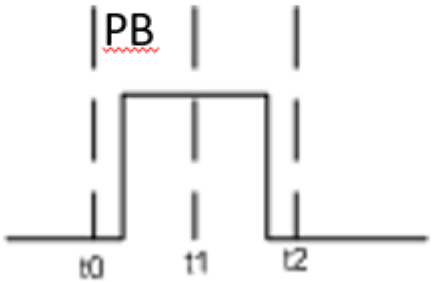
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PLC Program Scan Time (Cont.)

Scan Example



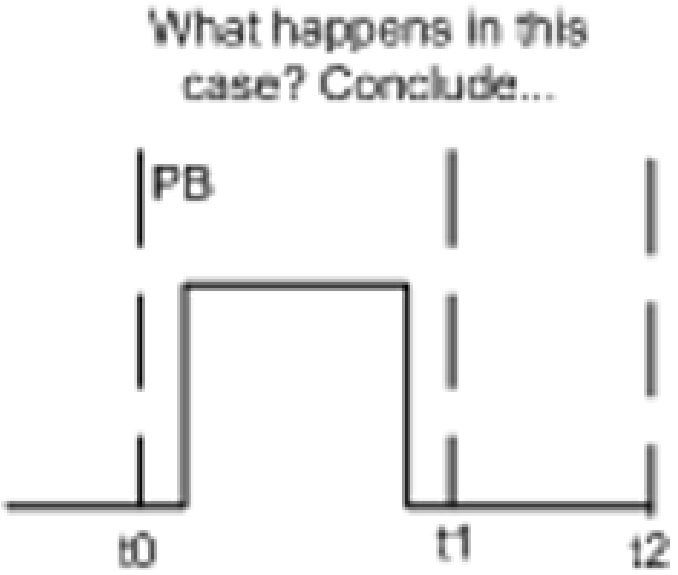
time	READ	EXECUTE	WRITE
t0	PB = 0 C = 0	C = 0	G = 0
t!	PB = 1 C = 0	C = 1	G = 1
t2	PB = 0 C = 1	C = 1	G = 1



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PLC Program Scan Time (Cont.)

- What happens in this case?



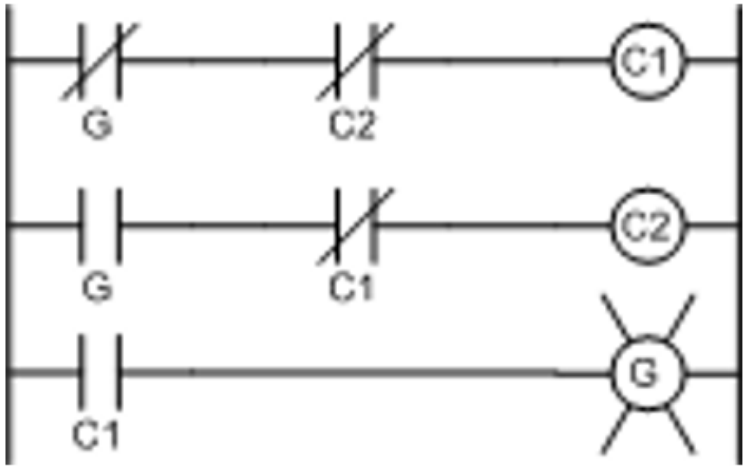
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Feedback and Scanning

- Notice the feedback of the previous ladder diagram. Feedback in combinational logic creates memory, like the previous latch, and creates “states” that depend on previous outputs/inputs.
- Consider the following feedback. What happens?

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Feedback and Scanning (Cont.)



time	READ	EXECUTE	WRITE
t0	G = 0 C1 = 0 C2 = 0	C1 = 1 C2 = 0	G = 1
t!	G = 1 C1 = 1 C2 = 0	C 1= 0 C2 = 1	G = 0
t2	G = 0 C1 = 0 C2 = 1	C1 = 0 C2 = 0	G = 0
t3	G = 0 C1 = 0 C2 = 0	C1 = 1 C2 = 0	G =1



The frequency of this oscillation depends on the PLC scan time. We do not have control on it.

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Allen-Bradley File Organization

Where are the files?

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Understanding the A-B Files

- Next slides talked about the files in the A&B PLC.
- It’s nice to know the file organization and where the files are.
- Indispensable for debugging and troubleshooting.
- It’s important to know that the data associated to each PLC device (contacts, timers, counters etc.) is stored in these files, and we will use them as we progress in the course.

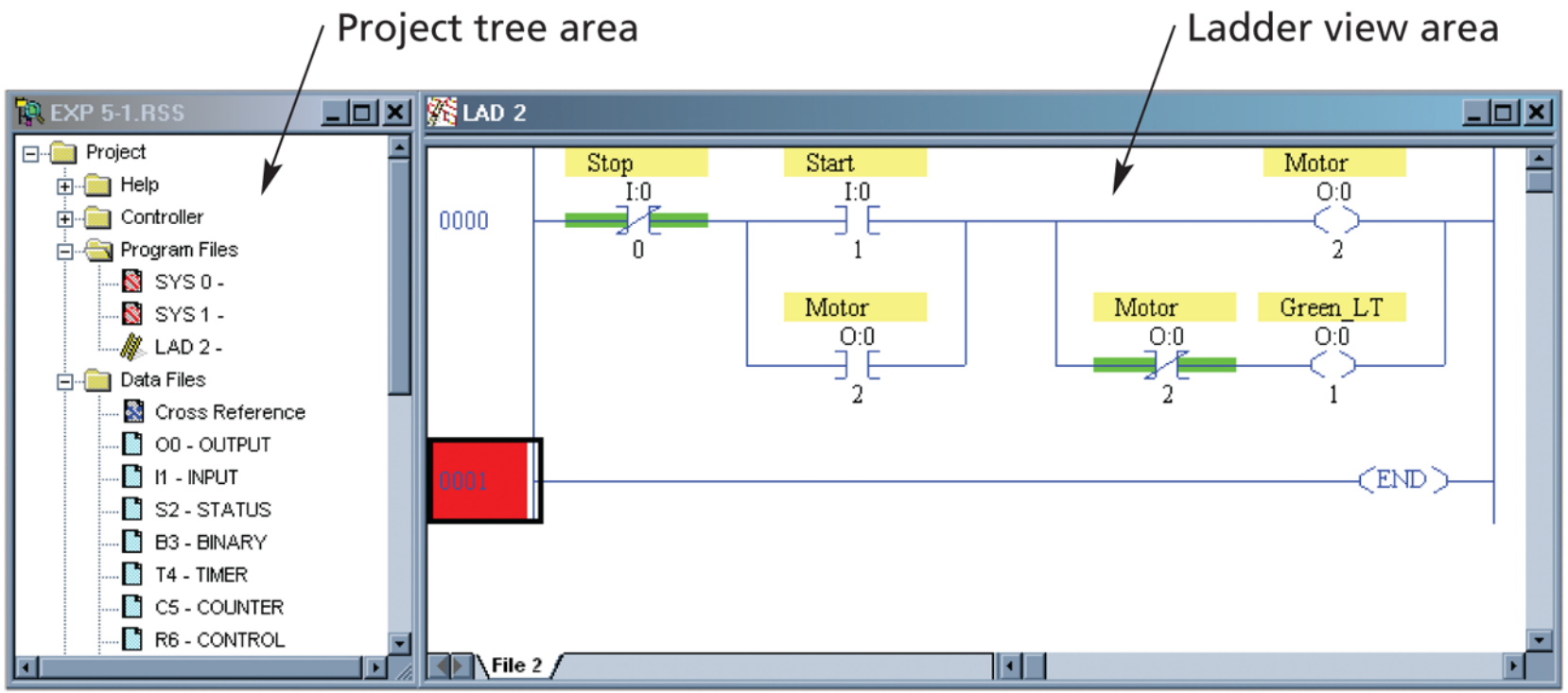
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Project Tree Area

- Divided into:
  - Program Files
  - Data Files
- Three program file icons in the Program Files area

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Project Tree Area (Cont.)



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Program Files

- SYS0 program file:
  - Used by PLC processor to keep track of its internal register settings
  - Not available to RSLogix 500 user
- SYS1 program file:
  - Reserved
  - Not available to RSLogix 500 user

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Main Program File (LAD2)

- Holds the main PLC ladder logic diagram
- PLC project:
  - May be more than one program file

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Subroutine Program Files

- May be more in addition to main program file
- Called on or accessed from main program file
- Selecting a main program file or a subroutine file in the project tree area will display its ladder logic diagram in ladder view area

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Data Files

- Input (I)
- Output (O)
- Status (S2)
- Timer (T4)
- Counter (C5)
- Control (R6)
- Integer (N7)
- Bit (B3 and B10)

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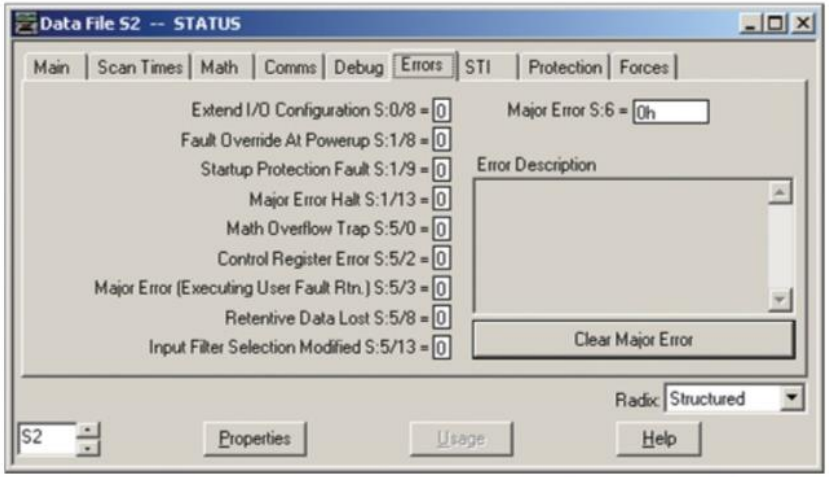
Data Files (Cont.)

- Input (I):
  - Inputs can be examined if open (XIO) or examined if closed (XIC)
- Output (O):
  - In the Allen-Bradley Fixed SLC 500 PLCs, eight output ports in module zero

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Status (S2)

- Holds flag or status bits of the ladder logic diagram
- If a central processing unit fault is generated in the PLC, click this icon
- Find fault indicator status bit
- Read the reason the message was generated
- Correct the program to solve the fault condition



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Figure 6-21. PLC status file.

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Timer (T4)

- Holds information related to the number of timers used in the ladder logic program
- Depending on the size of the PLC RAM, there could be timers with addresses from T4:0 to T4:255:
  - A PLC ladder diagram can have access to 256 timers
  - Number of timers is limited due to lack of enough RAM memory in the PLC

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Counter (C5)

- Holds information related to counters that can have any address from C5:0 to C5:255
- Number limited due to the lack of RAM memory in the PLC

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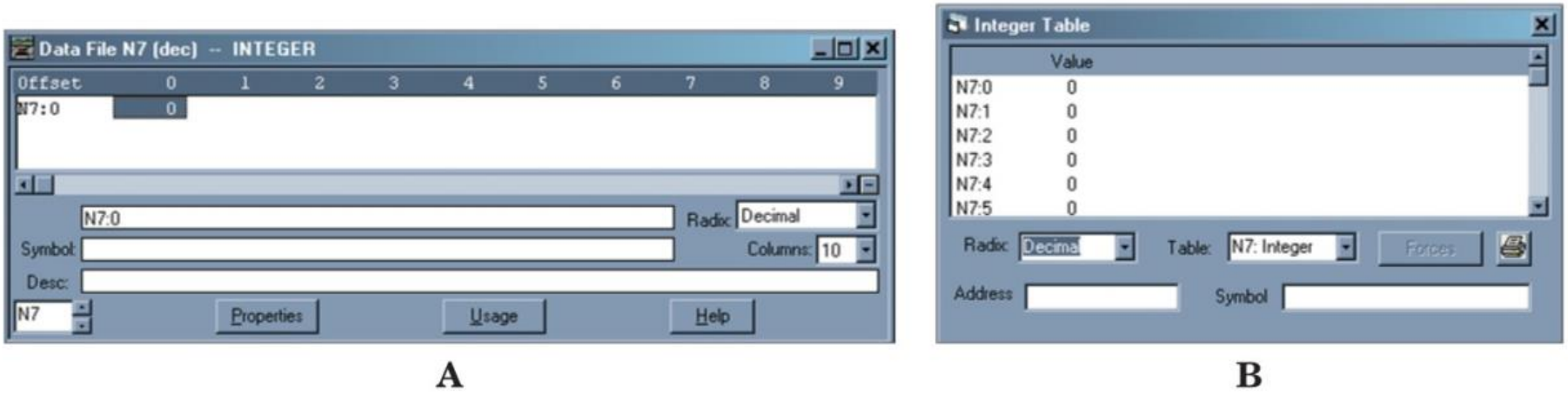
Control (R6)

- Control file (R6) holds 256 words (R6:0 to R6:255) used as control registers
- Control registers are used with sequencer and shift register instructions
- Registers R6:0 to R6:255 can hold valuable information related to the sequencer and shift register instructions

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Integer (N7)

- Holds 256 16-bit words (N7:0 to N7:255) that can be used as temporary storage integer registers



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Figure 6-22. Integer files. A—RSLogix 500 software from Rockwell International. B—LogixPro software from The Learning Pit.

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Bit (B3 and B10)

- Each holds 256 words (B3:0 to B3:255 and B10:0 to B10:255)
- Used by advance PLC instructions such as bit shift and sequencer instructions
- All Allen-Bradley SLC 500 series Programmable Logic Controllers have 16-bit registers

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